

**DONBOSCO INSTITUTE OF MANAGEMENT STUDIES AND
COMPUTER APPLICATIONS**

**A Project Synopsis
ON
COMPARE2SAVE
(2025-2026)**

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ABSTRACT:

The COMPARE2SAVE is a comprehensive, full-stack marketplace and local product discovery application designed to bridge the gap between neighborhood brick-and-mortar retail stores and modern e-commerce. The project is designed using Flask for the backend, SQLite for database operations, and modern frontend technologies with Three.js and Leaflet.js. The platform delivers dedicated dashboards with Role-Based Access Control (RBAC) for Customers, Shopkeepers, and Owner Admins.

Traditional shopping methods for local stores rely heavily on physical visits and manual price comparisons, which lead to high search friction and wasted travel time. The proposed system replaces these outdated routines with a location-aware digital catalog. Customers can verify securely using Twilio SMS OTP, search for products nearby, compare prices, and negotiate with shopkeepers via a real-time chat interface.

A core feature is the integration of Three.js for immersive 3D preview of products and an OpenAI-powered AI Shopping Assistant. Overall, the project aims to improve local trade and offer cross-platform accessibility as a PWA and Capacitor-based mobile app.

INTRODUCTION:

In recent years, the retail market has experienced a significant shift towards e-commerce. While large online retail provides convenience, it lacks instant local fulfillment, and neighborhood shopkeepers face immense difficulty making inventory visible to nearby consumers. Local consumers who need items immediately often walk from store to store, which is inefficient.

Traditional local commerce is fragmented. The COMPARE2SAVE application is developed to overcome these limitations by introducing a complete web-based solution that automates local marketplace activities.

The system provides an interactive platform enabling local businesses to catalog products, while giving neighborhood customers a live search engine to locate items and compare prices nearby.

OBJECTIVES:

- To develop a complete web-based local product discovery system using modern technologies
- To automate inventory management for local shopkeepers and reduce search overhead
- To implement secure authentication using Twilio SMS-based OTP verification
- To integrate 3D product visualization using Three.js
- To provide interactive maps using Leaflet.js to locate nearby shops
- To embed an OpenAI-powered AI Shopping Assistant for smart recommendations
- To establish real-time live chat communication between buyers and sellers
- To provide a scalable platform supporting PWA and mobile applications

EXISTING SYSTEM:

Existing local commerce operates primarily through manual routines. When consumers want to purchase a product immediately, they must visit local shops physically or make phone calls to inquire about availability and pricing.

These systems are difficult to navigate and highly inefficient. They lead to high search costs for consumers, frequent stock unavailability issues, and lost revenue for local shopkeepers who cannot publicize their stock.

Limitations:

- Manual store-visiting routines consume significant time and travel effort
- Lack of online visibility and digital cataloging tools for small shopkeepers
- No mechanism for real-time local product availability and pricing check
- Absence of intelligent technologies like AI recommendations and 3D visualization
- Inability for customers to chat and negotiate directly with nearby sellers online
- High reliance on distant e-commerce fulfillment instead of local immediate delivery

PROPOSED SYSTEM:

The proposed COMPARE2SAVE system provides an intelligent and automated solution for local product discovery and marketplace activities. The system is designed using Python (Flask), SQLite, HTML5, CSS3, JavaScript, Three.js, and Leaflet.js.

The application provides centralized dashboards for different roles. Customers can register, locate nearby products on interactive maps, and use an AI assistant for recommendations.

Advantages:

- Provides secure Twilio OTP authentication and authorization mechanisms
- Automates location-based store discovery using Leaflet.js
- Enables centralized management of inventory for shopkeepers
- Reduces manual searching and wasted travel time
- Improves operational efficiency and gives small businesses digital visibility
- Supports real-time chat and interactive 3D product previews

TECHNOLOGY USED:

Frontend:

- HTML5, CSS3 (Vanilla)
- JavaScript (ES6)
- Three.js (3D Viewer)
- Leaflet.js (Maps)

Backend:

- Python
- Flask, Flask-SQLAlchemy

Database:

- SQLite / PostgreSQL

Additional Technologies:

- Twilio API (SMS Auth)
- OpenAI API (Chatbot)
- Capacitor (Mobile Wrapper)

CONCLUSION:

The COMPARE2SAVE system provides a smart and efficient solution for modern local commerce. By integrating web technologies with interactive mapping, 3D visualization, and AI capabilities, the system simplifies product discovery, price comparison, and offline-to-online retail bridge.

The implementation of secure authentication ensures safety and verified interactions. The use of modern frameworks such as Flask, Three.js, and Leaflet.js ensures scalability, flexibility, and reliability.

Overall, the proposed system serves as a practical, scalable, and secure solution for the local retail ecosystem and can be enhanced in the future with advanced AR shopping experiences and localized delivery logistics.

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